

Results Evidence Map for *The Making of Vehicle Currencies*

Internal results packet. Not manuscript prose.

Purpose and Reading Order

This document reorganizes the current empirical evidence around research questions, not around the order in which the analyses were built. Table 1 is the master map: every empirical claim must point to one research question, one proxy, and one primary table. The tables use a JFE-style hierarchy: compact headline estimates in the main rows, conventional significance stars shown on estimates, p-values shown explicitly, fixed effects and clustering reported in notes, and weaker or conditional evidence labelled as such.

Interpretation Rules

1. Definitions are not propositions. Vehicle use is defined by intermediate-token use in indirect routes; the propositions are about emergence, persistence, rotation, architecture, and settlement design.
2. “Value” is not a primitive. Feasibility, depth/capacity, execution-cost advantage, settlement-transfer incidence, and LP-liquidity response are the measurable objects.
3. WETH, stablecoins, V3, and V4 are empirical test beds. They should not be written as the propositions themselves.
4. Conventional significance stars are shown on estimates (*, **, *** for 10%, 5%, and 1%). P-values are also reported directly so significance is not hidden behind stars. Weak, noisy, or pre-trending results are labelled in the table notes.

Table 1: Research questions, empirical objects, and evidence.

RQ	Research question	Unit	Main proxy	Evidence	Empirical answer
RQ1	When does one asset become the vehicle?	Endpoint pair x candidate vehicle x day.	Direct cost advantage, indirect-route availability, indirect-route depth.	Table 4	Actual vehicle share and future vehicle share rise with executable indirect-route economics.
RQ2	How does liquidity provision make a vehicle?	Token x day dynamic panel.	LP concentration, vehicle-linked liquidity, lagged vehicle share.	Table 5	Relative LP concentration and vehicle use are mutually persistent; absolute linked TVL moves differently.
RQ3	Why does vehicle status persist or get displaced?	Incumbent-challenger pair x day.	Lagged vehicle share and challenger route-cost edge.	Table 5	Incumbents persist, but large challenger cost edges predict share losses.
RQ4	When does vehicle status switch under stress?	Common-support stress-event days.	WETH-minus-stable vehicle share relative to a prior baseline.	Table 6	Stress rotates share away from WETH and toward stable vehicles on event days.
RQ5	How does architecture change vehicle formation?	Endpoint-pair event windows around V3.	Direct-route availability and no-direct/WETH-available cases.	Table 7	V3 expands direct-route opportunity and reduces no-direct dependence on WETH.
RQ6	How does settlement design change vehicle use?	Matched V3/V4 route cells and receipt-audited route units.	Intermediate-token transfer incidence and matched-cell V4 route use.	Table 8	V4 lowers physical intermediary transfers while vehicle use persists.
RQ7	Does a vehicle create common liquidity?	Pool x vehicle x day liquidity panel.	Leave-one-out vehicle liquidity factor.	Table 9	Vehicle-linked pools share a same-vehicle liquidity component beyond market liquidity.

Notes: The table is the organizing map for the results document. It separates the research question from the test bed, empirical proxy, and table that answers it.

Table 2: Vehicle-use measurement and exact-quote scope.

Panel	Quantity	Main estimate	Companion estimate	Interpretation
<i>Panel A. Vehicle-use denominators, 2026</i>				
	WETH	44.5% conditional BridgeShare	5.3% all-route bridge share	50.0% endpoint-pair coverage
	USDC	20.6% conditional BridgeShare	2.5% all-route bridge share	25.3% endpoint-pair coverage
	USDT	23.6% conditional BridgeShare	3.0% all-route bridge share	20.7% endpoint-pair coverage
<i>Panel B. Exact-quote scope</i>				
	Exact-quote covered share	78.9		Unified leg-volume share covered by V2/Sushi V2/V3 plus Balancer weighted quote extension
	Excluded Curve+Fluid share	16.2		Unified leg-volume share not covered by exact executable-depth quotes
	Excluded / covered ratio	0.205		Maximum exact-quote scope exposure relative to covered quoteable venues
	Excluded stable-leg share	75.3		Excluded venues are stablecoin-heavy, so route-cost claims must be scoped

Notes: BridgeShare is conditional on indirect routes. All-route bridge share and endpoint-pair coverage are reported so the conditional vehicle measure is not confused with a share of all DEX volume. Exact executable-depth route-cost results are scoped to covered quoteable venues.

Table 3: Variable construction and empirical proxies.

Variable	Level	Construction	Used for
Vehicle share	vehicle token x day	USD volume of indirect routes where token v is an intermediate divided by total indirect-route USD volume that day.	RQ1, RQ2, RQ3, RQ4
Direct cost advantage	endpoint pair x vehicle x day x trade size	(direct-route exact-quote output USD - indirect-route exact-quote output USD) / direct-route output USD. The candidate-day measure is the median fraction across common-support endpoint pairs at \$10k; positive values favor the direct route.	RQ1, RQ3, RQ5
Direct available	endpoint pair x day x trade size	Indicator that the exact-quote engine finds a direct executable route for the endpoint pair at the standard notional.	RQ1, RQ5
Indirect route available	endpoint pair x vehicle x day x trade size	Indicator that both legs of the indirect route through candidate vehicle v are exact-quote executable at the standard notional.	RQ1, RQ5
Vehicle-linked liquidity	vehicle token x day	Allocated USD TVL in valid Uniswap V3 daily-snapshot pools containing candidate v: full pool TVL when v is the only candidate side and half when both sides are candidates.	RQ2, RQ3, RQ6
LP concentration	vehicle token x day	VehicleLinkedLiquidity divided by total vehicle-candidate linked liquidity across the vehicle set that day.	RQ2, RQ3
Transfer incidence	matched route unit	Indicator that the transaction receipt contains at least one ERC-20 Transfer log matching the intermediate vehicle token.	RQ6
Vehicle liquidity factor	vehicle token x day	Leave-one-out average daily log-liquidity change among other pools linked to the same vehicle; paired with a leave-one-out market liquidity factor.	RQ7

Notes: All variables are generated from the reconstructed route and liquidity data. VehicleShare is conditional on indirect routed volume; all-route shares are used only as scope diagnostics.

Table 4: Vehicle formation: route economics, availability, and realized route choice.

	(1) Actual vehicle share	(2) Vehicle share $\tau = 7$	(3) Vehicle share $\tau = 30$
Direct cost adv. (fraction)	-14.6239*** p<0.001	-0.0187*** p<0.001	-0.0287*** p<0.001
Indirect-route avail.	19.1385*** p<0.001	0.1159*** p<0.001	0.1450*** p<0.001
Indirect-route depth	28.8502*** p<0.001		
LP concentration		0.2115*** p<0.001	
Lagged vehicle share			0.3466*** p<0.001
FE	Endpoint-pair x date	Token and date	Token and date
SE cluster	Date	Date	Date
Obs.	189,143	9,380	9,265

Notes: Cells report coefficients with p-values beneath them. Stars denote 10%, 5%, and 1% significance. DirectCostAdvantage is a direct-minus-indirect fraction of direct-route output, so a negative coefficient links stronger indirect-route performance to greater vehicle use.

Table 5: Liquidity provision, persistence, and challenger displacement.

	(1) Vehicle share $\tau = 7$	(2) LP conc. $\tau = 7$	(3) Log LP liq. $\tau = 7$	(4) Chg. LP conc. $\tau = 30$	(5) Chg. log LP liq. $\tau = 30$
<i>Panel A. Liquidity-route dynamics</i>					
LP concentration	0.2115*** p<0.001				
Lagged vehicle share		0.0150*** p<0.001	-0.2454*** p<0.001	0.6479 p=0.152	0.0231 p=0.105
Indirect-route avail.				2.1340*** p<0.001	0.0632*** p<0.001
Indirect-only avail.				-7.0424*** p<0.001	-0.1881*** p<0.001
FE	Token and date	Token and date	Token and date	Token and date	Token and date
SE cluster	Date	Date	Date	Date	Date
Obs.	9,380	9,380	9,380	9,265	9,265
<i>Panel B. Challenger displacement thresholds</i>					
Challenger j= incumbent				-2.80*** p<0.001	1,687
0-0.0025				-5.19*** p<0.001	86
0.0025-0.01				-6.72*** p<0.001	146
0.01-0.025				-9.50*** p<0.001	92
j0.025				-10.08*** p<0.001	193

Notes: Cells report coefficients with p-values beneath them. Panel A uses token and date fixed effects with date-clustered standard errors. Panel B reports incumbent share changes by challenger route-cost edge bins; the last column gives the number of incumbent days.

Table 6: Stress rotation inside common route opportunities.

	Events	(1) WETH or pre gap	(2) Stable or event gap	(3) Gap or post 1-7	(4) Direct or post 8-30	(5) Indir. vol. or note
Stress event day	20	-1.48 pp** p=0.018	1.48 pp** p=0.018	-2.96 pp** p=0.018	-0.25 pp p=0.679	0.51 log points*** p=0.005
<i>Panel B. WETH-minus-stable event-time gap</i>						
Gap change	20	-5.63** p=0.016	-26.96*** p=0.006	-11.16* p=0.077	-9.48** p=0.046	
<i>Panel C. Threshold and overlap sensitivity</i>						
All, 6%	91			-3.12 pp*** p<0.001		neg. share 67.0%
All, 8%	49			-2.95 pp*** p=0.001		neg. share 65.3%
All, 10%	26			-3.74 pp*** p=0.001		neg. share 65.4%
All, 12%	19			-3.18 pp** p=0.015		neg. share 63.2%
Top 20, 8%	18			-3.09 pp** p=0.024		neg. share 61.1%
Nonoverlap, 8%	33			-3.01 pp** p=0.020		neg. share 57.6%

Notes: Cells report event effects with p-values beneath them. Panel A is the same-day decomposition relative to a prior 28-day baseline. Panel B uses the same WETH-minus-stable gap in event-time windows; pre-movement means the result should be written as a common-support rotation, not a clean surprise-shock causal design.

Table 7: Architecture and direct-route opportunity.

	(1) No-direct WETH	(2) Q1 direct	(3) Q1 no-direct	(4) Q2 direct	(5) Q2 no-direct	(6) Q4 direct	(7) Q4 no-direct
Post V3	-25.81*** p<0.001	56.42*** p<0.001	-33.03*** p=0.007	51.36*** p<0.001	-47.51*** p<0.001	3.28* p=0.099	-1.91 p=0.336
Pair FE	yes	yes	yes	yes	yes	yes	yes
SE cluster	Pair	Pair	Pair	Pair	Pair	Pair	Pair
Obs.	1,822	1,122	1,122	1,434	1,434	8,914	8,914

Notes: Dependent variables are route-opportunity measures. Cells report post-V3 effects in percentage points with p-values beneath them. The table is scoped to route opportunity around V3 and should not be read as a broad causal launch effect for every V3 outcome.

Table 8: Settlement design, physical transfer incidence, and matched-cell route use.

	(1) All sizes	(2) Small	(3) Medium	(4) Large	(5) Log V4 route count	(6) Log V4 route vol.
V4 - V3 transfer inc.	-18.6 pp*** p<0.001	-39.4 pp*** p<0.001	-12.3 pp*** p=0.002	-3.2 pp* p=0.083		
V3 transfer inc.	100.0%	100.0%	100.0%	100.0%		
V4 transfer inc.	81.4%	60.6%	87.7%	96.8%		
<i>Panel B. Matched-cell route-use persistence</i>						
Log V3 route count					0.3674*** p<0.001	
Log V3 route volume						0.5819*** p<0.001
Week FE / vehicle FE					yes	yes
Week clusters					75	75

Notes: Columns (1)-(4) compare matched V3 and V4 route units; cells report transfer-incidence gaps with p-values beneath them. Columns (5)-(6) are matched endpoint-vehicle-week regressions of V4 route use on V3 route use.

Table 9: Common liquidity across pools linked to the same vehicle.

	(1) Full sample	(2) Stress interact.	(3) High vehicle use	(4) Low vehicle use	(5) Excl. top pools
Market liquidity factor	0.7063*** p<0.001	0.6987*** p<0.001	0.6572*** p<0.001	0.5276*** p<0.001	0.7062*** p<0.001
Vehicle liquidity factor	0.1626*** p<0.001	0.1586*** p<0.001	0.2174*** p<0.001	-0.0921 p=0.307	0.1670*** p<0.001
Vehicle factor x stress		0.1064** p=0.030			
Pool-vehicle FE	yes	yes	yes	yes	yes
SE cluster	Date	Date	Date	Date	Date
Obs.	685,685	685,685	664,218	21,467	679,228

Notes: Dependent variable is daily pool-level log liquidity change. Cells report coefficients with p-values beneath them. Regressions include pool-vehicle fixed effects and date-clustered standard errors. The vehicle factor is leave-one-out across other pools linked to the same vehicle.

Table 10: Specification registry for paper-facing tests.

Test	Unit	Sample	Outcome	Treatment / regressor	FE / SE	Main estimate	Interpretation
P1 availability	pair x day x size	quoteable venues	availability; direct cost adv.	WETH indirect route	pair-day aggregation	9,584 no-direct; direct cost adv. -0.0974/-0.4996/-0.9057	availability/thin-direct protection
P2 dynamics	token x day	vehicle candidates	share; LP	lagged LP concentration; lagged VehicleShare	token/date FE; date SE	LP- i share 0.227 (p <0.001); share- i LP conc. 0.014 (p <0.001); share- i log TVL 0.060 (p 0.054)	relative persistence; level sensitivity
P3 stress	event x pair set	WETH downside events	WETH-stable share	WETH stress day	event-level t-tests	-2.96 pp, p=0.018	same-day WETH-to-stable rotation
P4a V3	pair x month	V3 launch pairs	no-direct WETH	post-V3	pair FE; pair-clustered SE	-25.81 pp, p<0.001; pretrend p=0.922	direct-route opportunity expansion
P4b V4	route unit; token-week	matched V3/V4 cells	transfer inc.; LP response	V4; post x netting	matched tests; token/week FE	V4 gap -18.6 pp; log LP beta 2.132	netting lowers movement; LP response

Notes: This registry fixes the paper-facing unit, sample, outcome, treatment or regressor, inference convention, headline estimate, and bounded interpretation for each main test.